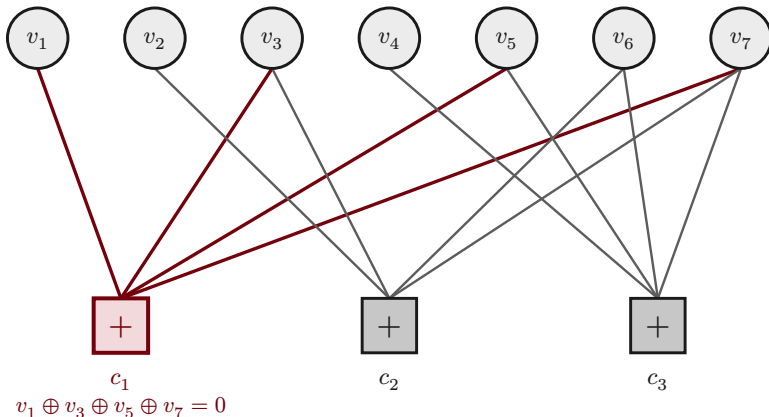


Tanner graph of an LDPC code

bi partite: bit nodes vs parity-check nodes; an edge wherever $H_{ij} = 1$ · sparse H = “low-density”

bit nodes
(variables, cols of H)



check nodes
(parity, rows of H)

H = parity-check matrix

$$\begin{matrix} c_1 \\ c_2 \\ c_3 \end{matrix} \begin{bmatrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 \\ \mathbf{1} & 0 & \mathbf{1} & 0 & \mathbf{1} & 0 & \mathbf{1} \\ 0 & \mathbf{1} & \mathbf{1} & 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}$$

$m = 3$ checks $\times$ $n = 7$ bits · edge $\Leftrightarrow H_{ij} = 1$

○ bit node v_j □+ check node c_i (parity) — incidence edge